ModelIt!

Student Activity Pack

Why Do Bridges Collapse?

Modeling Structural Forces, Material Fatigue, and Engineering Failure

Grade 11L1 | NGSS HS-PS2-1, HS-ETS1-2 | Stability and Change

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **What forces do you think act on a bridge when a truck drives across it? Try to name as many as you can.**

2. **Why do you think bridges are designed to hold much more weight than they normally carry — what could go wrong?**

3. **If a bridge looks perfectly fine from the outside, does that mean it is safe? Why or why not?**

4. **What do you think eventually causes an old bridge to collapse — one big event or something that builds up over time?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Applied Load | Structural Design Type | Maintenance Interval | Force Distribution | Fatigue Damage | Structural Integrity

EXTERNAL	INTERNAL

? Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Normal Service Life

Load: Standard traffic | Design: Steel truss | Maintenance: Every 2 years | 50-year simulation

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Deferred Maintenance

Load: Standard traffic | Maintenance: Every 10 years | 50-year simulation

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Overloading Event

Load: 150% of design capacity | Bridge at 70% integrity

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Corrosion Rate — The speed at which environmental exposure (salt, water, atmospheric chemicals) degrades structural steel and concrete — corrosion reduces the cross-sectional area of load-bearing members, amplifying the effect of fatigue cracks

Temperature Cycling — The daily and seasonal expansion and contraction of bridge materials due to temperature changes — thermal cycling creates additional stress cycles that contribute to fatigue, with extreme temperature ranges producing the most damage

Redundancy Level — The degree to which a bridge's structural design includes backup load paths — a redundant design allows forces to reroute around a failed member; a non-redundant design collapses when a single critical member fails

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

If engineers can calculate exactly how much force a bridge can handle, why do bridges still collapse — and what invisible process is weakening the structures we drive over every day? Use evidence from your simulation to support your answers.

1. **How does your model demonstrate that material fatigue is invisible yet dangerous — what makes it different from other types of structural damage?**

2. **What does the model reveal about the relationship between maintenance investment and bridge safety — is there a minimum inspection frequency below which risk increases dramatically?**

3. **Why do you think so many governments defer bridge maintenance despite knowing the consequences — what economic and political pressures contribute?**

4. **How could your model be used to design 'smart bridges' that monitor their own structural health and alert engineers to fatigue damage?**

5. **What limitations does your model have in predicting real-world bridge failures, and what additional factors would you need to include?**

8 STEM Challenge: Design a Bridge Monitoring and Maintenance Plan

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YOUR ENGINEERING MISSION

Design a comprehensive monitoring system and maintenance schedule for an aging bridge that maximizes safety while optimizing the use of limited inspection and repair budgets.

THE SCENARIO

A state department of transportation manages 1,200 bridges, 40% of which are over 50 years old and classified as 'structurally deficient.' The annual maintenance budget is only $50 million — not enough to repair all bridges simultaneously. Your structural engineering team must design a risk-based inspection and maintenance plan that prioritizes the most dangerous bridges, optimizes inspection frequency based on age and traffic volume, and prevents catastrophic failures within budget constraints.

Engineering Design Thinking

1. WHAT SENSOR TECHNOLOGIES COULD PROVIDE CONTINUOUS FATIGUE MONITORING BETWEEN SCHEDULED INSPECTIONS?

2. HOW WILL YOU BALANCE THE COST OF EARLY INTERVENTION (CHEAPER PER BRIDGE BUT SERVING MANY BRIDGES) VERSUS EMERGENCY REPAIR (EXPENSIVE PER BRIDGE BUT FEWER BRIDGES)?

3. AT WHAT STRUCTURAL INTEGRITY THRESHOLD SHOULD A BRIDGE BE CLOSED TO HEAVY TRAFFIC OR SHUT DOWN ENTIRELY?

4. HOW WILL YOUR PLAN ACCOUNT FOR CLIMATE CHANGE IMPACTS — INCREASED FLOODING, TEMPERATURE CYCLING, AND EXTREME WEATHER — ON BRIDGE FATIGUE RATES?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Structural Engineers design bridges, buildings, and other load-bearing structures, performing force analysis and fatigue calculations to ensure safety over decades of service, earning $70,000–$140,000/year. Bridge Inspectors conduct hands-on assessments of structural integrity, using ultrasonic testing, magnetic particle inspection, and visual analysis to detect fatigue cracks, earning $55,000–$95,000/year. Infrastructure Asset Managers optimize maintenance budgets for large portfolios of bridges and roads, earning $80,000–$130,000/year.